

The empirical recurrence for OEIS A295713: recovering a conjecture that is not written down

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Abstract

OEIS A295713 records “Empirical recurrence of order 76”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 81$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 76, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 76 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295713 is “Number of $n \times 5$ 0..1 arrays with each 1 horizontally or vertically adjacent to 0, 1, 2 or 4 1s.”. It has offset 1 and begins

32, 768, 17364, 396571, 9056598, 206899873, 4725935539, 107948235688, ...

The entry states:

Empirical recurrence of order 76 (see link above)

As of the “Last modified” line on the live entry (Nov 26 2017 09:26:33, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 81$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 81$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 162$ exact terms of the model returns order 76 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{76} c_i a(n-i),$$

c_1	14	c_{20}	458701013	c_{39}	672461515	c_{58}	1122793
c_2	129	c_{21}	154672072	c_{40}	329758274	c_{59}	796948
c_3	1372	c_{22}	−730260466	c_{41}	−456834175	c_{60}	1023431
c_4	5835	c_{23}	−677578762	c_{42}	−739932935	c_{61}	−971164
c_5	20012	c_{24}	327765364	c_{43}	−470078080	c_{62}	−397340
c_6	13382	c_{25}	620571286	c_{44}	−386308083	c_{63}	56333
c_7	−58973	c_{26}	−83053098	c_{45}	−110955781	c_{64}	−6433
c_8	−322011	c_{27}	−235430402	c_{46}	267606253	c_{65}	−99887
c_9	−423535	c_{28}	145213197	c_{47}	301359394	c_{66}	11001
c_{10}	−572250	c_{29}	168752852	c_{48}	152929276	c_{67}	41379
c_{11}	−5165779	c_{30}	−479226142	c_{49}	114079950	c_{68}	−18686
c_{12}	374804	c_{31}	−358987122	c_{50}	3661407	c_{69}	3897
c_{13}	9504494	c_{32}	186404006	c_{51}	−89481028	c_{70}	2155
c_{14}	−6414083	c_{33}	140315959	c_{52}	−63868198	c_{71}	−1179
c_{15}	−9294710	c_{34}	−321714798	c_{53}	−14274651	c_{72}	479
c_{16}	77049393	c_{35}	−416860500	c_{54}	−13060175	c_{73}	−197
c_{17}	46661079	c_{36}	154951913	c_{55}	−4756214	c_{74}	19
c_{18}	−183561096	c_{37}	677678227	c_{56}	10989923	c_{75}	12
c_{19}	−17238896	c_{38}	858108408	c_{57}	7383276	c_{76}	−4

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 76, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $76 + 4$ terms removed returns order 76 as well, so $\deg q_{\min} = 76 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $76 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 76 that this sequence satisfies — and that family turns out to have exactly one member.

References

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